CampusChain

Smart Campus Financial Ecosystem

PROBLEM STATEMENT:	**Unified Blockchain-Powered Campus Financial OS for Institutions**
COLLEGE NAME:	**A.P. Shah Institute of Technology, Thane**
TEAM NAME:	**ViTi**
TEAM MEMBERS:	**Viresh Nair | Tanay Bhatkar | [Team ViTi]**

Problem Statement

Identifying gaps in how campus finances are currently managed

Clearly Defined Problem

- Campus financial systems operate in fragmented silos — fee payments,
- vendor transactions, event ticketing, and peer-to-peer transfers are all
- handled by separate, disconnected tools with no unified oversight.

Domain & Context

- Higher Education | FinTech | Blockchain Technology
- Affects students, faculty, vendors, and administrative staff daily.

Key Challenges

- Cash dependency on campus — no digital-first unified payment layer.
- No transparent audit trail for inter-campus financial transactions.
- Manual fee collection is slow, error-prone, and hard to reconcile.
- Students have no visibility into spending patterns or balances.

Existing Inefficiencies & Gaps

- Current UPI/bank solutions lack campus-specific role-based controls.
- No mechanism for admin-controlled token minting or disbursement.
- Event ticketing involves paper-based or third-party tools (Insider, etc.).
- Zero fraud detection or anomaly alerting in any existing campus system.

Impact on Stakeholders

- Students: financial opacity, no instant transfer capability, cash hassles.
- Admins: no real-time dashboards, manual reconciliation burden.
- Vendors: delayed settlements, no digital transaction records.
- Institution: compliance risk, no audit trail, fragmented reporting.

Proposed Solution

A unified, blockchain-backed campus financial operating system

CampusChain is a full-stack, role-aware financial platform that runs on top of a Supabase-powered PostgreSQL backend with a blockchain-inspired immutable ledger. It allows admins to mint CampusCoin (CC), students to pay fees, transfer funds P2P, buy event tickets, and vendors to accept digital QR payments — all in one dashboard.

High-Level System Architecture

Frontend
(Next.js + React)

→

API Layer
(Next.js Routes)

→

Supabase
(PostgreSQL + Auth)

→

Blockchain Ledger
(SHA-256 Blocks)

Component Interaction Flow

1. User authenticates → JWT issued → stored in localStorage.
2. All API calls carry the Bearer token → server validates via jsonwebtoken.
3. Transactions are written to the Ledger table + a SHA-256 Block is mined.
4. Supabase Realtime broadcasts balance changes → dashboard updates live.
5. Fraud engine checks rules post-transaction → flags inserted into fraud_alerts.
6. EmailJS sends transaction receipt to both sender and receiver.

Approach / Methodology

Step-by-step engineering strategy adopted to build CampusChain

Step 1 — Database-First Design

Modeled all entities (users, ledger, blocks, events, tickets, fraud_alerts) in PostgreSQL via Supabase. Row-Level Security policies lock every resource to the correct role (student / admin / vendor).

Step 2 — Blockchain-Inspired Immutability

Every financial transaction generates a unique SHA-256 tx_id and a corresponding Block record (block_number, prev_hash, merkle_root). This creates an append-only audit chain that cannot be altered without breaking hash continuity.

Step 3 — JWT Role-Based Auth

Custom /api/auth/login endpoint issues JWTs with embedded {id, role}. All API routes validate the token server-side using jsonwebtoken, enforcing least-privilege access control per user type.

Step 4 — Modular API Architecture

Each major domain (wallet, ledger, events, fraud, analytics) has its own /api/* route. This separation of concerns simplifies testing, debugging, and future microservice extraction.

Step 5 — Rule-Based Fraud Detection

A fraud engine evaluates every outbound transaction against 4 heuristic rules: (a) 3× average transaction amount, (b) >5 txns in 10 minutes, (c) first-time large transfer to unknown recipient, (d) suspicious round amounts. Flags are stored in fraud_alerts and surfaced on the admin dashboard.

Step 6 — Real-Time UX

Supabase Realtime subscriptions push balance changes and incoming transfers to connected clients instantly. Toast notifications alert users without a page refresh, keeping the experience fast and interactive.

Tech Stack (Proposed & Implemented)

Every technology was chosen for its real-world production readiness

Frontend

Next.js 16

React-based framework with server-side rendering & file-based routing (Pages Router)

React 19

Component-driven UI with hooks for state, effects, and real-time updates

Tailwind CSS 4

Utility-first CSS framework for a responsive, dark-mode-ready design system

Lucide React Icons

Utilities

Recharts

Declarative charting library used in the Analytics & Spending pages

Node.js crypto

SHA-256 hashing for transaction IDs, block hashes, and Merkle root generation

QRCode

Server-side QR code generation for vendor payment codes and event tickets

EmailJS

Client-side email SDK for sending transaction receipts to both parties instantly

date-fns

Lightweight, tree-shakeable date formatting for transaction timestamps

Backend & DB

Next.js API Routes

Serverless backend endpoints: /api/auth, /api/wallet, /api/events, /api/fraud…

Supabase (PostgreSQL)

Cloud-hosted relational DB with Row-Level Security, Realtime subscriptions & admin SDK

Supabase Realtime

WebSocket-based pub/sub that pushes live balance and ledger updates to clients

Security & Auth

jsonwebtoken (JWT)

Stateless authentication — tokens embed user id + role; validated on every request

bcryptjs

Salted password hashing (cost=10) for all user password storage and verification

Expected Impact & Innovation

What makes CampusChain unique compared to existing campus payment tools

🔗 Blockchain Audit Trail

Unlike any campus ERP, every transaction is linked in an immutable SHA-256 block chain. Admins can verify the integrity of every payment since the genesis block.

🛡️ Built-In Fraud Intelligence

A rule-based fraud detection engine runs on every outbound transaction, checking velocity, amount anomalies, and unknown recipients — a feature absent in all standard campus payment apps.

⚡ Real-Time Balance & Notifications

Supabase Realtime delivers instant push updates to wallets. Students see incoming money appear live with toast alerts — no page refresh required, no delay.

🎟️ NFT-Like Event Ticketing

Event tickets are on-chain tokens with a unique ticket_token UUID, linked to a transaction ID in the ledger. Tickets cannot be duplicated or forged.

📊 Spending Analytics Dashboard

Category-level spending breakdown (fees, P2P, vendor, events) with trend charts powered by Recharts — giving students financial self-awareness not seen before.

🏆 Gamified Leaderboard

A community leaderboard ranks students by token activity, encouraging responsible financial participation and engagement with campus digital services.

Feasibility & Scalability

Designed for real-world deployment from day one

Technical Feasibility

- Fully open-source stack (Next.js + Supabase) — zero licensing cost.
- Deployed on Vercel (frontend) + Supabase Cloud (backend) in < 30 minutes.
- All components are stateless and horizontally scalable.
- JWT authentication works without sessions — scales to millions of concurrent users.

Operational Feasibility

- Institution admin creates accounts, mints CC, and monitors from one dashboard.
- Students onboard via email registration — no app download required (browser-first).
- EmailJS receipts keep all parties informed without additional infrastructure.
- QR-based vendor payments need only a smartphone camera — no POS hardware.

Scalability Plan

- Supabase supports connection pooling (PgBouncer) and read replicas for high load.
- Vercel Edge networking auto-scales API routes globally with zero config.
- Blockchain table can be sharded by academic year (block ranges per term).
- Multi-campus rollout: each campus gets an isolated Supabase project/schema.

Future Expansion

- Mobile app (React Native) reusing the same API layer — no backend changes.
- Smart contract integration (Ethereum/Solana) for on-chain token minting.
- AI-powered fraud scoring (ML model replacing rule-based engine).
- Integration with national payment rails (UPI, NEFT) for token top-ups.
- OpenID Connect / Aadhaar eKYC for verified student identity assurance.

Thank You

CampusChain — Smart Campus Financial Ecosystem

Team ViTi | A.P. Shah Institute of Technology

Built with ● Next.js ● Supabase ● Blockchain ● React